

AntennaMaster User Guide

RF Coverage Simulator — Complete Reference

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Everything you can do with AntennaMaster, every setting it exposes, and what each one means. All numbers in this guide are taken directly from the code.

Companion documents:

- `README.md` — quick overview and architecture
- `INSTALL_GUIDE.md` — cross-platform install & launch (Windows/macOS/Linux)
- `docs/CAPABILITIES.md` — condensed capability/capacity matrix
- `docs/ROADMAP.md` — five-layer capability model & delivered phases
- `VISION_ARCHITECTURE.md` — 3D digital twin, live telemetry, drone LiDAR
- `SaaS_ARCHITECTURE.md` — multi-tenant/SaaS internals
- `http://localhost:8000/docs` — live interactive OpenAPI reference

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1. Getting started

One-command start

```
./install.sh # macOS/Linux: scan the host, install missing runtimes, build
./launch.sh # boot both servers, wait for health, open the browser
```

On Windows use `install.ps1` / `launch.ps1` (PowerShell). The installer auto-resolves Python 3.10+, Node 18+ and any build tools; see `INSTALL_GUIDE.md` for the full matrix. For development you can

still use:

```
./start.sh # installs deps if needed, starts backend :8000 + frontend :3000
./start.sh --check # runs the full QA gate first (backend tests, benchmarks, frontend tests)
```

Manual start

Backend (Python ≥ 3.11):

```
cd backend
pip install -r requirements.txt
uvicorn app.main:app --port 8000
```

Frontend (Node ≥ 18 ; proxies `/api/*` to the backend via `BACKEND_URL`, default `http://localhost:8000`):

```
cd frontend
npm install
npm run build && npm start # or: npm run dev
```

Open **`http://localhost:3000`**. The REST API is self-documenting at **`http://localhost:8000/docs`**.

Health probes

- GET `/api/health` — liveness.
- GET `/api/ready` — readiness (data directory writable, DEM cache state).

First study in 30 seconds

- 1 Click the map once to place the **TX** (transmitter), click again for **RX**.
- 2 The terrain profile, line-of-sight verdict and Fresnel clearance appear automatically — no DXF or account needed anywhere on Earth (SRTM 30 m).
- 3 Pick a technology preset (e.g. *LTE 800*) in the **Radio study** panel to turn the profile into a full link budget with received power per sample.

2. The planner (main screen)

Language, Simple Mode & the guided tour

- **Language** — the header has an **EN / FR** switcher. English and French ship out of the box; your choice is remembered. (Adding a language is just another `locales/<code>/common.json` file.)
- **Simple ↔ Expert** — the header toggle. **Simple Mode** hides the RF numbers (dBi, MHz, propagation model) and instead asks “*what are you trying to connect?*” — pick an outcome (connect two buildings, Wi-Fi for a vehicle fleet, handheld radios across a site, private 4G/5G, IoT sensors, mobile coverage, long-distance backhaul) and the right technology and settings are applied for you. **Expert** exposes every control.
- **Guided tour** — first-time visitors get a short walkthrough (placing points → choosing a technology → uploading a DXF → reading the heatmap). Replay it any time with the  button.
- **Glossary tooltips** — every technical field has an  icon. Hover, focus or tap it for a plain-language, equation-free explanation (also translated).

Map & providers

Setting	What it does
Base layer picker	6 built-in providers: OpenStreetMap, OpenTopoMap, Carto Light, Carto Dark, Esri World Imagery, Esri World Topo
Custom XYZ URL	Any {z}/{x}/{y} tile template (private tile servers, orthophotos...)
Theme toggle	auto (follows OS) → dark → light; persisted
Language	English / French; persisted

The map view (center/zoom) and the whole session (endpoints, study settings, DXF reference) persist in `localStorage` and are restored on reload.

Placing TX and RX

- **Click** the map: first click sets TX, second sets RX; further clicks move RX.
- **Type exact coordinates:** the TX/RX lat/lon fields accept free typing (values apply when they parse; you can clear the box while editing).
- **GPS:** *Use my location* sets the TX from the browser's geolocation (with a visible error if permission is denied).
- **Swap:** exchanges TX and RX in one click.
- **Search:** enter `lat`, `lon` directly or a place name (geocoded through OpenStreetMap Nominatim — needs internet).

Antenna heights

`TX height` and `RX height` are meters **above ground level** at each end. Leave a technology preset selected and heights are pre-filled with realistic values (e.g. 30 m mast / 1.5 m handheld for GSM).

Profile settings

Setting	Range	Default	Meaning
Samples	16–2,048	256	points along the geodesic path
k-factor	0.1–10	1.333	effective-earth radius factor; 4/3 = standard atmosphere, 0.7 ≈ sub-refraction worst case, 2+ ≈ ducting
Foliage depth	0–400 m	0	vegetation crossed by the path (Weissberger model)
Rain rate	0–150 mm/h	0	rain fade (ITU-R P.838/P.530), matters above ~7 GHz
Clutter %loc	0–99.9	0 (off)	ITU-R P.2108 statistical man-made clutter; 50 = median urban, 90 = conservative
Surface model	on/off	off	sample a DSM (buildings as obstructions); visible only when <code>AM_DSM_URL</code> is configured

The profile recomputes automatically (debounced ~350 ms) whenever an input changes. **Export CSV** downloads the full per-sample table with the same study parameters applied.

3. DXF terrain workflow

A DXF is a *local override*: AntennaMaster works globally on SRTM alone, and a georeferenced DXF patches higher-resolution relief inside its footprint.

Step 1 — Upload

Drag-and-drop (or browse) a `.dxf` file into the wizard. Cap: **100 MB** (`AM_MAX_DXF_MB`). Damaged files go through ezdxf's recovery loader.

The upload response includes **auto-detected hints**: coordinate magnitudes are analyzed to suggest a probable UTM zone or a feet-based drawing, and the wizard pre-fills the georeferencing mode accordingly.

Step 2 — Pick terrain layers

Every layer is listed with its entity types, point count and Z-range, plus a "terrain-likeness" heuristic that pre-selects plausible relief layers. Elevation is extracted from:

- `POINT` entities (survey points)
- `LWPOLYLINE` with an `elevation` attribute (contour lines)
- `POLYLINE` — 2D/3D polylines, polyface meshes, polygon meshes
- `3DFACE` and `MESH`
- `TEXT` / `MTEXT` spot heights (regex-parsed, `EL=` / `H=` prefixes supported)

Duplicate XY points are de-duplicated; the scattered cloud is gridded with `scipy.griddata` (linear, nearest-neighbour fallback at the hull fringe) to a density-aware grid of at most 400×400 cells.

Step 3 — Georeference (three modes)

Mode	You provide	Best when
Known CRS	EPSG code or PROJ string (e.g. <code>EPSG:32633</code>)	the drawing is in a documented projected CRS (UTM, Lambert, state plane)
Control points	2–3 pairs of DXF X/Y ↔ real Lat/Lon	you can identify surveyed corners/landmarks in the drawing
Origin + bearing	anchor Lat/Lon, true bearing of the +Y axis, unit (m/ft/yd/cm), optional origin offset	you know where the drawing origin sits and how it is rotated

Control points solve a least-squares 2D **Helmert** transform (scale + rotation + translation) in a local plane and report **per-point and RMS residuals in meters** — if a residual is large, that control point is wrong. An independent `z_scale` converts vertical units (e.g. `0.3048` for elevations in feet).

Validation & fusion

After georeferencing, the mean DXF elevation is cross-checked against mean SRTM elevation over the same footprint. A difference above **50 m** (`AM_VALIDATION_DIFF_M`) raises a strict warning — the classic causes are a feet/meters mix-up or a wrong CRS.

In profiles and coverage, the DXF wins inside its bounding box; across a **3-grid-cell feathered band** (`AM_FEATHER_CELLS`) the DXF and SRTM surfaces are cross-faded so the seam never creates a false diffraction edge. Every profile sample carries its provenance (`srtm` / `blend` / `dxf`), which the chart colors blue/orange.

The map shows the DXF **footprint polygon** and a semi-transparent **hillshade overlay** (hypsothetic tint × analytic hillshade) so you can see exactly what terrain the DXF contributes. DXF state is persisted server-side and rebuilt automatically after a restart.

4. Point-to-point radio studies

Select a **technology preset** in the Radio study panel (see §13 for all 23). Every preset field can be overridden.

Study settings

Setting	Meaning
Technology	preset bundle: frequency, powers, gains, sensitivity, heights, best-fit model
Model	override the propagation model (§14)
Environment	model-specific: urban/suburban/open (Hata), +metropolitan (COST-231), los/nlos (3GPP)
Frequency	MHz; out-of-validity values are clamped with an explicit API warning
TX power / TX gain / RX gain / losses	link-budget terms (dBm / dBi / dBi / dB)
RX sensitivity	threshold for the margin verdict (dBm)
Fade margin	extra dB subtracted before the verdict — plan ~5.5 dB for 90 % area confidence, ~8 dB for 95 %
Foliage depth	Weissberger vegetation loss (0–400 m)
Rain rate	ITU-R P.838/P.530 rain fade (0–150 mm/h)
Clutter %loc	ITU-R P.2108 statistical man-made clutter: 0 = off, 50 = median urban, 90 = conservative planning (0.5–67 GHz; distance-dependent)
Surface model	sample a DSM (buildings/canopy as obstructions) instead of bare terrain — shown when the server has <code>AM_DSM_URL</code> configured
k-factor	earth curvature (0.1–10, default 4/3)

What you get

- Per-sample **received power** along the path, path-loss / diffraction split.
- **LOS verdict** and **first Fresnel zone** clearance (60 % rule).
- **Worst obstruction**: location, height and knife-edge v of the dominant obstacle, marked on the chart.
- **Margin vs sensitivity** verdict for the chosen technology.
- Terrain diffraction is always computed with **Deygout multi-knife-edge** (up to 3 edges, ITU-R P.526) on the k-curved *fused* profile — on top of whichever empirical model is active.

Channel-aware presets (Private LTE/5G/IoT) derive sensitivity from thermal noise: $-174 + 10 \cdot \log_{10}(\text{BW}) + \text{NF} + \text{SINR}$, and add MIMO gain to the budget. With clutter enabled the study shows a separate **Clutter (P.2108)** line in the loss breakdown (as do foliage, rain and gases).

Refraction reliability (microwave)

Every profile response also carries a **dual-k reliability check** under `rf.refraction` — the standard microwave test that a dependable hop clears **100 % of the first Fresnel zone at $k = 4/3$** (standard atmosphere) **and 60 % at $k = 2/3$** (sub-refraction worst case). Fields: `f1_ratio_k43`, `f1_ratio_k23`, and a boolean `reliable`. A hop that passes at 4/3 but fails at 2/3 will drop out during anomalous propagation.

Antenna height optimizer

GET `/api/terrain/optimize-heights` returns the **minimum TX and RX height** (by bisection, holding the other end fixed) that achieves bare line of sight and the 60 %-first-Fresnel rule — the "how tall does the mast need to be?" question, answered without trial and error. `null` means the criterion is unreachable within the height cap (default 120 m).

4a. Batch receiver qualification

For fixed-wireless / WISP work — qualifying many subscriber addresses against one tower — the **Batch receivers** panel (and POST `/api/rf/batch`) takes a list of up to 200 locations (`name,lat,lon` per line) and returns each one's distance, RX power, margin, served verdict, LOS and Fresnel clearance in a single call. The panel renders a sortable table, colours served/unserved, lets you click a row to drop the RX pin there, and exports the whole set as **CSV** (`?format=csv`) for a CRM or spreadsheet. It honours the same foliage / rain / clutter / DSM settings as the profile study.

4b. Best-site search

POST `/api/rf/site-search` ranks an $n \times n$ grid of candidate TX positions over a bounding box (2×2 to 7×7) by coarse served-area fraction — the "where should the mast go?" question. It uses low-resolution sweeps to stay interactive; re-run the winning coordinate through full coverage. Both batch and site-search are Pro-tier features in SaaS mode.

5. Area coverage simulation

Run coverage sweeps a polar grid outward from the TX and paints a margin-classed raster on the map.

Parameters

Setting	Range	Default	Notes
Radius	0.1–150 km	10	simulation extent
Radials	36–720	180	angular resolution
Steps per radial	20–400	100	range resolution
Raster size	128–1,024 px	512	output image resolution
Antenna	omni / sector / uploaded MSI pattern	omni	see §7
Azimuth	0–360°	—	sector/MSI pointing
H-beamwidth	5–360°	65	parametric sector (-3 dB)
V-beamwidth	1–90°	10	parametric vertical cut
Downtilt	–10...+20°	0	mechanical tilt (MSI electrical tilt is read from the file)
Shadow-fade margin	0–30 dB	0	subtracts a log-normal fade allowance from every pixel — design to 90/95 % area
Foliage / rain / clutter / k-factor	as in §4		applied per-step
Surface model	on/off	off	needs <code>AM_DSM_URL</code> ; buildings become obstructions

Output

- **Raster overlay** with 5 margin classes (strong → marginal), transparent where unserved; legend included.
- **Served-area fraction**, area-weighted (annulus-correct, not pixel-count).
- Peak/edge statistics: TX ground elevation, strongest RX power, radius.
- **Exports**: PNG (georeferenced bounds), **GeoTIFF** (EPSG:4326, imports directly into QGIS / ArcGIS / Atoll / Pathloss) and **KMZ** (Google Earth GroundOverlay).

Per-step terrain diffraction uses the vectorized single-strongest-edge kernel, numerically verified against the float64 reference to < 0.05 dB. A 10 km, 180×100 sweep completes in ~2–3 s with a warm DEM cache.

6. Multi-site best-server maps

Enterprise-tier feature. Build a site list and composite them:

- 1 Configure a TX (position, antenna, azimuth, downtilt) and click **Add current TX as site** — repeat for up to **8 sites**.
- 2 **Best-server map** runs a full coverage simulation per site and paints each pixel in the color of the strongest serving site (colorblind-safe palette).
- 3 Per-site **best-server shares** and a combined served fraction are shown; the raster is exportable like single-site coverage.

SINR / interference analysis

Every multi-site run (2+ sites) also computes a **co-channel SINR map**: per pixel, $SINR = S / (I + N)$ where S is the best server, I the linear sum of every other site heard there (worst-case frequency-reuse-1 — all sites on the same carrier) and N the thermal noise floor ($-174 + 10 \cdot \log_{10}(BW) + NF$, taken from the preset's channel parameters or the request's `bandwidth_mhz / noise_figure_db`; a 10 MHz / 7 dB default is used — and flagged in a warning — when neither is available).

The panel gains a **Best server / SINR** view toggle plus three statistics: mean SINR over the served area, % of area at ≥ 6 dB (comfortable MCS), and the cell-edge fraction below 0 dB. The SINR raster has its own 5-class green-to-amber legend and PNG URL. Set `interference: false` in the API call to skip it.

Multi-site accepts the same physics settings as single-site (radials capped at 360, steps at 200 per site; raster up to 1,024 px on the union bbox).

7. Antenna patterns

Three ways to model the TX antenna:

- 1 **Omni** (default) — gain applied uniformly.
- 2 **Parametric 3GPP sector** — set azimuth, horizontal beamwidth (–3 dB), vertical beamwidth and downtilt. Pattern: $12 \cdot (\Delta/BW)^2$ capped at 25 dB front-to-back horizontally and a 20 dB floor vertically.

- 3 Measured MSI Planet file** — upload a `.msi` / `.pln` / `.txt` pattern (≤ 2 MB). Gain is converted dBd→dBi (+2.15) automatically; electrical tilt read from the header; attenuation is the sum of the H and V cuts. Uploaded patterns are listed with gain and -3 dB beamwidths and are private to your account when signed in.

8. Indoor & underground studio

Open **Indoor** / **Underground** for the three tabs of studies no DEM-based tool can run. Pro-tier feature in SaaS mode.

Tab 1 — Floor plan / mine map coverage

Uses a DXF as *structure*, not relief — no georeferencing needed; everything runs in drawing coordinates.

- 1 Upload a floor plan (walls read from LINE, LWPOLYLINE, POLYLINE, ARC/CIRCLE — arcs tessellated at 15°).
- 2 Assign a **material per layer** — layer names are auto-guessed against the 12-material library (§15); mark decorative layers as *Ignore*.
- 3 Set the **unit scale** (meters per drawing unit), frequency (or a preset), TX/RX heights (default 2.5 m / 1.2 m) and the grid resolution (50–400 px, default 200).
- 4 **Multi-floor** (optional): set *Floors crossed* (0–30) when the TX sits N storeys away from the floor you mapped. The engine adds the COST-231 floor-penetration term $L_f \cdot n^{((n+2)/(n+1)) - 0.46}$ — per-slab loss configurable (default 18.3 dB $\approx$ concrete slab), saturating with n because energy increasingly leaks via stairwells/windows — and stretches the 3D distance by the storey height. The P.1238 fallback honors the floor count too.
- 5 **Click the rendered plan** to place the TX — or type exact plan X/Y.

The engine computes FSPL over 3D distance **plus the sum of wall crossings** (COST-231 multi-wall; exact vectorized segment-intersection tests) for every grid cell, and composites a heatmap with the walls drawn on top. Statistics: served %, RX power dynamic range. If **no walls** are found on the selected layers, it automatically falls back to **ITU-R P.1238** site-general indoor loss and says so in a warning. Export: heatmap PNG.

Tab 2 — Tunnel / mine gallery link

Emslie waveguide model — reproduces why UHF outranges VHF underground.

Setting	Range	Default
Frequency	> 0 MHz	446
Tunnel width / height	0.5–30 m	4 / 3
Length	10–50,000 m	2,000
Wall preset	concrete, rock, coal, limestone, salt (ϵ_r 4–7)	rock
Polarization	horizontal / vertical	horizontal
Roughness	0–1 m	0.1
Tilt	0–10°	0
TX power / gains / losses / sensitivity	free	30 dBm / 6 dBi / 0 / 0 / –100 dBm

Output: RX-power-vs-distance chart, attenuation in dB/m, the modal **breakpoint** (before it the direct ray dominates; after it the guided mode), and maximum range vs sensitivity.

Tab 3 — Through-the-earth (TTE)

VLF magnetic-loop link through conductive ground (mine emergency comms).

Setting	Range	Default
Frequency	10 Hz–1 MHz	5,000 Hz
Depth	1–2,000 m	100
Ground preset	dry rock (0.001 S/m) → wet clay (0.1 S/m)	average soil (0.01)
TX power / system gain / sensitivity	free	30 dBm / 20 dB / –130 dBm

Output: **skin depth**, ground attenuation vs near-field spreading split, total loss, margin and verdict.

9. Reading the profile chart

- **Terrain fill** — blue = SRTM samples, orange = DXF samples (blend zone shades between). Elevations are drawn on the k-curved earth.
- **Green line** — line of sight TX→RX.
- **Dashed line** — lower edge of the first Fresnel zone; terrain intruding past it costs diffraction loss even with visual LOS.
- **Red dot** — worst obstruction (highest knife-edge v).
- **Tooltip** — per-sample elevation, data source, LOS height, Fresnel edge and (with a technology selected) received power in dBm.
- Charts longer than 512 samples are downsampled for rendering with a peak-preserving algorithm (obstacle peaks are never smoothed away); exports always contain every sample.

10. Accounts, roles & dashboards

Accounts are optional in open mode (see `AM_SAAS_MODE` in §17). Sign in from the planner header. Registration asks for email, password, organization, **role** and starting tier.

Role	Dashboard	Built for
Manager (Enterprise IT)	/dashboard — Command Center	project portfolio CRUD, cost/ROI estimator KPIs, plan management, white-label logo upload, org audit log
Field technician	/field — Tactical View	forced dark theme, live GPS follow with spot elevation checks, one-tap technology presets that seed the planner
Pre-sales	/pitch — Pitch Interface	A/B scenario comparison with async jobs + progress bars, ROI calculator (payback, 5-yr net), executive PDF

All three link back to the planner; the role only picks the default landing experience — nothing is locked away by role (tiers do that, §12).

Security properties you get for free: PBKDF2-SHA256 passwords (200k iterations), 30-day bearer tokens with logout revocation, login lockout (8 failures / 15 min), resource ownership (your DXFs and antenna patterns are yours), org-scoped audit trail.

11. Projects, sharing & reports

Projects

Save as project snapshots the entire planner state (endpoints, study settings, DXF reference, sites) into a named workspace. From the dashboard (or the API) you can list, rename, duplicate and delete projects; opening `/?project=ID` restores one exactly. Quotas per tier: 3 / 25 / unlimited.

Sharing

Share creates a public read-only token URL for a project (the token is stripped from responses of anyone but the owner). Anyone with the link can view — revoke by deleting the project.

PDF reports

`POST /api/saas/report.pdf` (or the *Executive PDF* button in the Pitch dashboard) renders a branded report: link-budget matrix (with environmental loss lines), the terrain profile chart, coverage raster and a CAPEX/OPEX/5-year-TCO bill of materials. Enterprise accounts with an uploaded logo (PNG/JPEG ≤ 1 MB) get **white-label** branding.

Cost estimator & hardware BOM

`GET /api/saas/costs?technology=...&sites=...` — per-technology BOM (radio, antenna, install, licensing...) with CAPEX, yearly OPEX and 5-year TCO used by the dashboard and pitch ROI views. The Command Center's **Bill of materials (CSV)** link (`GET /api/saas/bom.csv`) downloads the same BOM with line items scaled to the fleet plus CAPEX/OPEX/TCO summary rows — the procurement deliverable for a purchase order.

Field use & offline (PWA)

AntennaMaster is an installable progressive web app. A service worker caches the app shell, every map tile you've viewed, and your last API results, so the tool keeps working **off-grid** — the decisive requirement for open-pit, underground and remote last-mile sites. The **Tactical view** (`/field`) shows a live **online / offline** indicator; when offline it falls back to cached tiles and results. Add it to a phone's home screen from the browser's "Install app" / "Add to Home Screen" menu. (The service worker activates in production builds only.)

Async jobs

Long simulations can run as background jobs (`POST /api/saas/coverage/async` → `GET /api/saas/jobs/{id}` with live progress %). At most **4 jobs** run concurrently; a 5th returns HTTP 429 — retry after one finishes.

12. Subscription tiers

	Basic \$0/mo	Pro \$79/mo	Enterprise \$299/mo
Global SRTM terrain, all map providers	✓	✓	✓
Consumer presets (Wi-Fi, PMR, broadcast, cellular, IoT)	✓	✓	✓
Saved projects	3	25	unlimited
DXF terrain fusion	—	✓	✓
PtP backhaul preset (18 GHz) with rain & gas	—	✓	✓
Indoor / underground studio	—	✓	✓
PDF reports	—	✓	✓ (white-label)

	Basic \$0/mo	Pro \$79/mo	Enterprise \$299/mo
Private LTE B48 / NR n77 / LTE-M presets	—	—	✓
Multi-site best-server	—	—	✓
API tokens	—	—	✓

Feature keys (for the API): `srtm_terrain` , `wifi_presets` → basic; `dxs_fusion` , `ptp_backhaul` , `pdf_export` , `indoor_studio` → pro; `private_networks` , `multi_site` , `api_access` , `white_label` → enterprise.

In **open mode** (default, `AM_SAAS_MODE` unset) nothing is gated and no account is required. In **SaaS mode** a gated call without the right tier returns HTTP **402** with the feature name and required tier. Tier changes are applied by a billing webhook (`POST /api/auth/tier` with `X-Billing-Secret`), not by the user directly.

13. Reference: technology presets

All fields overridable per study. Operators can merge extra band plans from `DATA_DIR/technologies.json` without code changes (new keys must carry the full field set).

Key	Preset	MHz	Default model	Env	TX dBm	TXg dBi	RXg dBi	Losses dB	Sens dBm	TX h / RX h (m)
<code>gsm900</code>	GSM 900 (2G)	945	Okumura-Hata	suburban	43	15	0	3	−102	30 / 1.5
<code>gsm1800</code>	GSM 1800/DCS (2G)	1842	COST-231	urban	43	17	0	3	−102	30 / 1.5
<code>umts900</code>	UMTS 900 (3G)	942.5	Okumura-Hata	suburban	43	15	0	3	−117	30 / 1.5
<code>umts2100</code>	UMTS 2100 (3G)	2140	38.901 UMa	nlos	43	18	0	3	−117	30 / 1.5
<code>lte800</code>	LTE 800/B20 (4G)	806	Okumura-Hata	suburban	46	15	0	3	−105	30 / 1.5
<code>lte1800</code>	LTE 1800/B3 (4G)	1842.5	COST-231	urban	46	17	0	3	−103	30 / 1.5
<code>lte2600</code>	LTE 2600/B7 (4G)	2655	38.901 UMa	nlos	46	18	0	3	−100	30 / 1.5
<code>nr700</code>	5G NR n28 700 MHz	758	Okumura-Hata	suburban	46	15	0	3	−105	30 / 1.5
<code>nr3500</code>	5G NR n78 3.5 GHz	3550	38.901 UMa	nlos	49	24	0	3	−100	25 / 1.5
<code>nr28000</code>	5G NR n257 28 GHz mmWave	28000	38.901 UMi	nlos	40	30	10	2	−95	10 / 1.5
<code>pmr446</code>	PMR446 handheld	446.1	Okumura-Hata	open	27	0	0	0	−119	30 / 1.5
<code>tetra400</code>	TETRA 400 (PPDR)	420	Okumura-Hata	suburban	44	9	0	2	−112	40 / 1.5
<code>vhf150</code>	VHF land mobile 150	155	Okumura-Hata	open	44	3	0	1.5	−116	40 / 1.5
<code>fm100</code>	FM broadcast 87-108	100	FSPL	open	60	6	0	1.5	−90	100 / 10
<code>dvbt600</code>	DVB-T/T2 UHF 600	600	Okumura-Hata	open	63	10	12	3	−84	150 / 10
<code>wifi2400</code>	Wi-Fi 2.4 GHz outdoor	2442	38.901 UMi	nlos	20	8	2	1	−82	10 / 1.5
<code>wifi5800</code>	Wi-Fi 5.8 GHz PtMP	5800	38.901 UMi	los	23	16	14	1	−80	20 / 5
<code>lora868</code>	LoRaWAN 868 (IoT)	868	Okumura-Hata	suburban	14	3	0	0.5	−137	25 / 1.5
<code>ptp18000</code> †	Microwave PtP 18 GHz	18000	FSPL	los	20	38	38	2	−70	30 / 30

Key	Preset	MHz	Default model	Env	TX dBm	TXg dBi	RXg dBi	Losses dB	Sens dBm	TX h / RX h (m)
private_lte_b48 ‡	Private LTE B48/CBRS	3625	38.901 UMa	nlos	40	15	0	1	−102 *	15 / 1.5
private_nr_n77 ‡	Private 5G NR n77 100 MHz	3900	38.901 UMa	nlos	47	24	0	1	−95 *	15 / 1.5
private_lte_iiot ‡	Private LTE-M/NB-IoT 1.4 MHz	3625	38.901 UMa	nlos	40	15	0	1	−120 *	15 / 1.5
custom	Custom study	446	FSPL	open	30	0	0	0	−100	20 / 1.5

† Pro tier in SaaS mode · ‡ Enterprise tier in SaaS mode · * channel-aware: sensitivity recomputed from bandwidth (20 / 100 / 1.4 MHz), noise figure (7 dB) and target SINR (−3 / −3 / −6 dB); MIMO gain (3 / 6 / 0 dB) added to the budget.

14. Reference: propagation models

Key	Model	Valid frequency	Environments	Typical use
fspl	Free space (ITU-R P.525)	1 MHz–300 GHz	—	baseline, microwave PtP
okumura_hata	Okumura-Hata	150–1,500 MHz	urban, suburban, open	GSM900, TETRA, PMR, FM/TV
cost231_hata	COST-231 Hata	1,500–2,000 MHz	urban, metropolitan, suburban, open	GSM1800/DCS, UMTS2100
tr38901_rma	3GPP TR 38.901 RMa	0.5–30 GHz	los, nlos	4G/5G rural macro
tr38901_uma	3GPP TR 38.901 UMa	0.5–100 GHz	los, nlos	LTE / 5G urban macro
tr38901_umi	3GPP TR 38.901 UMi	0.5–100 GHz	los, nlos	small cells, mmWave street

All models are floor-bounded by FSPL. Out-of-validity inputs are clamped and a warning is attached to the response. **Deygout multi-knife-edge terrain diffraction** (≤ 3 edges, ITU-R P.526) is always added on top, computed on the k-curved fused profile.

Environmental add-ons (any model): Weissberger foliage (0–400 m), ITU-R P.838-3/P.530 rain fade (specific attenuation $\times$ effective path length), ITU-R P.676-style gaseous absorption (oxygen 60 GHz / water-vapor 22 GHz lines — automatic, significant only for high-frequency PtP), and **ITU-R P.2108 §3.2 statistical clutter** — the man-made land-use loss the DEM cannot see, at a settable percentage of locations (50 = median, 90 = conservative; defined 0.5–67 GHz, sub-500 MHz requests use the 0.5 GHz curve with a warning).

Indoor/underground models: COST-231 multi-wall, ITU-R P.1238, Emslie tunnel waveguide, TTE skin-depth — see §8.

15. Reference: materials & presets

Wall materials (dB per crossing, log-frequency interpolated)

Material	900 MHz	2.4 GHz	5.8 GHz
Drywall / plasterboard	2	3	4
Wood / door	3	4	6

Material	900 MHz	2.4 GHz	5.8 GHz
Glass (plain)	1.5	2	3
Glass (low-E coated)	10	12	15
Brick wall	6	8	12
Concrete 20 cm	10	13	20
Reinforced concrete 30 cm+	18	23	32
Metal / shielding	26	30	35
Rock pillar (mine)	25	35	45
Earth / backfill	30	45	60
Elevator / machinery	20	25	30
Ignore (decorative layer)	0	0	0

Tunnel wall presets (relative permittivity ϵ_r)

concrete 6.0 · hard rock (granite) 5.0 · coal 4.0 · limestone 7.0 · salt 4.5

Earth conductivity presets (S/m)

dry rock/granite 0.001 · limestone (karst) 0.005 · average soil 0.01 · coal measures 0.02 · wet soil/clay 0.1

16. Reference: every setting

Consolidated list of every user-adjustable parameter with its valid range.

Path & profile ([GET /api/terrain/profile](#) , planner sidebar)

Parameter	Range	Default
TX/RX lat, lon	± 90 / ± 180	—
TX height / RX height (m AGL)	≥ 0	20 / 10
Samples	16–2,048	256
k-factor	0.1–10	4/3
technology / model / environment	see §13–14	—
freq_mhz	> 0	preset (else 446)
tx_power_dbm, tx_gain_dbi, rx_gain_dbi, losses_db, rx_sensitivity_dbm	free	preset
foliage_depth_m	0–400	0
rain_rate_mm_h	0–150	0
clutter_pct	0–99.9 (0 = off)	0
surface	true/false (needs AM_DSM_URL)	false
dxid	georeferenced DXF	none (SRTM only)

Coverage ([POST /api/rf/coverage](#))

Parameter	Range	Default
radius_km	0.1–150	10

Parameter	Range	Default
n_radials	36-720	180
n_steps	20-400	100
raster_px	128-1,024	512
antenna_azimuth_deg	0-360	omni
antenna_beamwidth_deg	5-360	65
vertical_beamwidth_deg	1-90	10
downtilt_deg	-10-+20	0
antenna_id	uploaded MSI id	—
shadow_margin_db	0-30	0
k_factor	0.1-10	4/3
foliage_depth_m / rain_rate_mm_h	0-400 / 0-150	0 / 0
clutter_pct	0-99.9	0
surface	true/false	false
+ all link-budget overrides from §4		

Multi-site ([POST /api/rf/coverage/multi](#))

Same as coverage, except: sites 1-8 (each with lat/lon, azimuth, downtilt, optional antenna), n_radials 36-360 (default 120), n_steps 20-200 (default 80), raster_px default 768. SINR analysis: interference (default true), bandwidth_mhz (0-400, else preset/10), noise_figure_db (0-20, else preset/7).

Batch receivers ([POST /api/rf/batch](#))

lat / lon (TX), receivers 1-200 (each lat , lon , optional name , rx_height_m), technology , dxf_id , surface , k_factor , foliage_depth_m , rain_rate_mm_h , clutter_pct , and the same link-budget overrides as coverage. ?format=csv for a CSV download.

Best-site search ([POST /api/rf/site-search](#))

south / west / north / east bbox, grid_n 2-7 (default 5), technology , radius_km 0.1-50, shadow_margin_db , clutter_pct , k_factor , dxf_id , surface , and TX-side budget overrides.

Height optimizer ([GET /api/terrain/optimize-heights](#))

lat1 / lon1 / lat2 / lon2 , tx_height_m , rx_height_m , freq_mhz or technology , k_factor , max_height_m (1-500, default 120), dxf_id , surface .

Indoor coverage ([POST /api/indoor/coverage](#))

Parameter	Range	Default
unit_scale (m per drawing unit)	> 0	1
layer→material map	\$15	auto-guessed
tx_x, tx_y (drawing coords)	in plan	click
tx_height_m / rx_height_m	> 0	2.5 / 1.2
floors_crossed	0-30	0
floor_height_m	2-6	3
floor_loss_db	0-40	18.3
grid_px	50-400	200

Parameter	Range	Default
freq_mhz or technology	> 0	preset
tx_power_dbm etc.	free	preset

Tunnel ([GET /api/indoor/tunnel](#))

freq > 0 (446) · width 0.5–30 m (4) · height 0.5–30 m (3) · length 10–50,000 m (2,000) · wall preset (rock) · polarization h/v · roughness 0–1 m (0.1) · tilt 0–10° (0) · budget terms free.

TTE ([GET /api/indoor/tte](#))

freq 10 Hz–1 MHz (5,000) · depth 1–2,000 m (100) · earth preset (average_soil) · tx_power_dbm (30) · system_gain_db (20) · rx_sensitivity_dbm (–130).

Georeferencing ([POST /api/dxf/{id}/georeference](#))

mode `known_crs` (epsg/proj string) | `control_points` (2–3 pairs) | `origin_bearing` (lat, lon, bearing 0–360, unit scale, offsets) · `z_scale` free (default follows horizontal) · layer selection.

UI-only settings

Base layer + custom XYZ URL · theme (auto/dark/light) · fade margin · search · GPS · swap ends · sites list · project save/load.

17. Server configuration (environment variables)

All backend behavior is tunable via `AM_*` environment variables — no code edits needed.

Variable	Default	Purpose
<code>AM_DATA_DIR</code>	backend/data	root for DEM cache, DXF store, results, SQLite DB
<code>AM_DEM_URL</code>	AWS Terrarium template	any Terrarium-encoded XYZ elevation source
<code>AM_DEM_ZOOM</code>	12 (≈ 38 m/px)	DEM tile zoom; higher = finer + more tiles
<code>AM_DSM_URL</code>	unset	optional Terrarium-encoded surface model source (buildings/canopy); enables <code>surface=true</code> on profiles and coverage
<code>AM_BASEMAP_URL</code>	OSM	source for the local base-map tile server (/api/basemap) used for offline maps
<code>AM_DEM_CACHE_MB</code>	2048	disk budget for the tile cache (LRU-evicted)
<code>AM_FEATHER_CELLS</code>	3.0	width of the DXF↔SRTM blending band, in grid cells
<code>AM_VALIDATION_DIFF_M</code>	50.0	mean-elevation mismatch that triggers the strict warning
<code>AM_MAX_DXF_MB</code>	100	DXF upload cap
<code>AM_CORS_ORIGINS</code>	*	comma-separated allowed origins
<code>AM_SAAS_MODE</code>	unset (open)	<code>1</code> = enforce accounts, tiers and quotas
<code>AM_BILLING_SECRET</code>	unset	shared secret required by the tier-change webhook in SaaS mode

Frontend: `BACKEND_URL` (default `http://localhost:8000`) — where the Next.js server proxies `/api/*`.

Operator band plans: drop a `technologies.json` into `AM_DATA_DIR` to merge custom presets (each new key needs the complete field set; existing keys are never overridden).

18. Limits & capacities

Resource	Limit
DXF upload	100 MB (configurable)
MSI antenna file	2 MB
White-label logo	1 MB (PNG/JPEG)
DXF terrain grid	$\leq 400 \times 400$ cells
Profile samples	$\leq 2,048$
Coverage sweep	≤ 720 radials $\times$ 400 steps, ≤ 150 km radius
Coverage raster	$\leq 1,024$ px
Sites per composite	8
Indoor grid	≤ 400 px
Concurrent async jobs	4 (429 beyond)
Stored raster results	last 200 (disk, auto-pruned; any worker serves any result)
DEM RAM cache	2,000 decoded tiles
Saved projects	3 / 25 / ∞ by tier
Session tokens	30-day TTL, revoked on logout
Login lockout	8 failures / 15 minutes
Benchmark gates (CI)	every endpoint scenario < 5 s, < 1 GB — see <code>QA_BENCHMARK_REPORT.md</code>

Verified by 100 backend test cases + 10 frontend tests (`./start.sh --check`).

19. Troubleshooting

Symptom	Cause & fix
"DXF mean elevation differs from SRTM by X m" warning	almost always feet-vs-meters (set unit scale / <code>z_scale</code> 0.3048) or a wrong EPSG code — check Helmert residuals
Large Helmert residuals (m)	one control point is misidentified; re-pick it
HTTP 402 on an endpoint	feature is tier-gated in SaaS mode — response says which tier unlocks it
HTTP 429 on async coverage	4 jobs already running; wait and retry
HTTP 429 on login	lockout after 8 failed attempts — wait 15 min
HTTP 502 from terrain endpoints	DEM tile source unreachable — check internet / <code>AM_DEM_URL</code>
Place search does nothing	Nominatim geocoding needs internet; <code>lat</code> , <code>lon</code> entry always works offline
Coverage looks too optimistic	add a shadow-fade margin (5.5 dB $\approx$ 90 %, 8 dB $\approx$ 95 % area), enable P.2108 clutter for built-up areas, and check the environment setting (nlos vs los)
422 "No surface model configured"	<code>surface=true</code> needs <code>AM_DSM_URL</code> pointing at a Terrarium-encoded DSM tile source
SINR warning about assumed receiver	the preset has no channel parameters — pass <code>bandwidth_mhz</code> / <code>noise_figure_db</code> in the multi-site request

Symptom	Cause & fix
Indoor heatmap ignores walls	selected layers had no line entities — the P.1238 fallback warning tells you; check layer selection and material mapping
Profile chart looks flat under the LOS line	you may be zoomed on a long path — hover the worst-obstruction dot; the chart is k-curved, terrain drops with distance
Model warning about frequency	you're outside the model's validity range — the value was clamped; pick a suitable model from §14

Known modeling boundaries (by design, documented in `docs/CAPABILITIES.md`): median empirical models + knife-edge diffraction (no ITM/P.1546/P.452 — Deygout + the Hata/38.901 family covers the same planning use cases); clutter is statistical (ITU-R P.2108), not a per-pixel land-use database; SINR assumes worst-case co-channel reuse-1 (no frequency plan / scheduler); multi-floor is a penetration term, not per-floor wall maps; building obstruction needs a user-supplied DSM tile source (`AM_DSM_URL`).

20. Advanced modules, 3D digital twin & live operations

Beyond predicting coverage, AntennaMaster now *designs*, *certifies* and *operates*. Open **Advanced studies** from the planner for the link-level tools; the 3D view, Live Operations and LiDAR are described below. Full endpoint reference: `VISION_ARCHITECTURE.md` and the OpenAPI page.

Two-way "talk-back" (land-mobile radio)

A radio system is limited by the *weaker* of two directions. The two-way tool computes **talk-out** (base → portable) and **talk-in** (portable → base), grades each in **DAQ** (Delivered Audio Quality, TIA-4046) and reports the limiting direction and the reliable talk-back area. It models portable **body loss** and a **penetration class** (on-street / in-vehicle / in-building / underground), and a **repeater-cascade** solver returns the spacing and count for continuous corridor coverage.

Endpoint	Purpose
POST <code>/api/rf/two-way/link</code>	bidirectional link verdict (talk-out/talk-in, DAQ, limiting direction)
POST <code>/api/rf/two-way/coverage</code>	area study: talk-out / talk-in / reliable-both served fractions
POST <code>/api/rf/two-way/repeater-cascade</code>	repeater count & spacing for a corridor

Leaky feeder & distributed antennas (metro / mine)

The real way tunnels are covered — a **radiating (leaky) coaxial cable** or a **distributed antenna system**. The tool models cable longitudinal + coupling loss, solves **inline-amplifier spacing**, adds bend loss, and reports served length and worst uncovered gap; the DAS designer turns the Emslie single-antenna reach into an antenna count and spacing. POST `/api/indoor/leaky-feeder` · GET `/api/indoor/tunnel-das` .

Automated AP placement (campus / warehouse)

The inverse of the heatmap: given a floor plan and a coverage (and optional capacity) target, POST `/api/indoor/auto-place` returns **how many** access points, **where**, and on **which channel** — greedy maximum-coverage placement over the same multi-wall engine, a graph-colour channel plan (2.4 / 5 / 6 GHz), a users × throughput capacity floor and a −67 dBm roaming-overlap check.

RF-exposure / EMF compliance

`POST /api/rf/compliance` computes **ICNIRP** or **FCC OET-65** public and occupational exclusion-zone distances and the exposure ratio at a given distance — the permitting gate before an antenna is switched on.

Longley-Rice (ITM) with a reliability quantile

`GET /api/terrain/itm` adds an **Irregular Terrain Model** path loss that delivers a *reliability quantile* (fraction of time and situations), not just a median — the statistical model empirical curves lack. It combines the validated Deygout diffraction with the Longley-Rice terrain-roughness statistic and the ITM time/situation variability.

Drive-test calibration

`POST /api/rf/calibrate` fits an offset (and optional distance-slope) correction from **measured RSSI** versus prediction and reports the RMS error before and after — turning predictions into site-tuned, trusted predictions.

AI design copilot

`POST /api/copilot/analyze/link` runs the profile **and** the height optimizer, then explains the result as ranked, actionable findings with a **quantified fix** (raise the mast to X m, add Y dB, change band, add a repeater). It is deterministic and works offline; an optional Claude narrator adds prose when an API key is present. `GET /api/copilot/tools` publishes a machine-readable tool catalog so an external agent can drive the simulator.

3D digital twin (CesiumJS)

A seamless **2D / 3D** toggle on the map renders the fused SRTM+DXF terrain in a WebGL globe — fed by the platform's own heightmap tiles (`GET /api/terrain/heightmap/{z}/{x}/{y}.bin` , no Cesium Ion key, offline-capable). It draws a glowing 3D **Fresnel tube** along the true line of sight, red markers where terrain slices into the Fresnel zone, and drapes the coverage heatmap over the 3D relief.

Live Operations (digital-twin telemetry)

The **Live Operations** dashboard (`/live`) ingests real-time asset positions from fleet-management or IoT feeds (`POST /api/telemetry/ingest` , `WebSocket /api/telemetry/ws`) and streams the live twin over Server-Sent Events (`/api/telemetry/stream`). Each asset is correlated against the RF prediction: it flashes **yellow** on entering a predicted **dead zone**, and an **RF-disconnect** event is logged (with whether the last position was in a dead zone) when it stops transmitting. Bind the prediction with `POST /api/telemetry/coverage-context` .

Drone LiDAR / point-cloud ingestion

`POST /api/lidar/upload` ingests a `.las` / `.laz` survey and rasterises it into a **Digital Surface Model** that overrides the statistical clutter, so diffraction is computed against the **real surveyed buildings, trees and machinery**. `GET /api/lidar/{id}/profile` returns a surveyed-surface-vs-bare-terrain comparison (validated: a 50 m building raised a link's modelled diffraction from 40 dB to 114 dB).